DAY 3

Shotgun Metagenomics & Assembly

QC & host removal • Kraken2 • MetaPhlAn4 • MEGAHIT/SPAdes • QUAST • SEQKIT

Amplicon vs Shotgun Metagenomics

Feature	Amplicon (16S)	Shotgun WGS
What is sequenced	One marker gene (16S rRNA)	All DNA in sample
PCR step	Yes — amplifies target gene	No — random fragmentation
Taxonomic resolution	Genus level (typically)	Species / strain level
Functional data	None	Full gene complement
Genome reconstruction	Not possible	Yes — MAGs
Novel organism detection	Limited by primers	Yes — unbiased
Cost (per sample)	~$30–80	~$100–300
Data size (per sample)	~10 MB	~1–10 GB
Bioinformatics complexity	Moderate	High
Best for	Community profiling, large cohorts	Functional potential, MAGs, discovery

Today we focus on shotgun — the foundation for Day 4 (MAGs) and Day 5 (Functional Analysis).

Shotgun Metagenomics Workflow

01 Raw Reads QC

FASTQC + MultiQC

Check read quality, adapter contamination, base composition, and GC content.

02 Trimming

fastp / Trimmomatic

Remove adapters, low-quality bases; filter short reads.

03 Host Removal

Bowtie2 / BWA

Map reads to host genome (human/plant), keep unmapped reads.

04 Read-based Taxonomy

Kraken2 + Bracken

Fast k-mer classification. Bracken estimates species-level abundances.

05 De Novo Assembly

MEGAHIT / SPAdes

Assemble reads into contigs using de Bruijn graphs.

06 Assembly QC

QUAST + SEQKIT

Evaluate contig statistics: N50, length distribution, genome completeness.

Today's Dataset

Category	This course's data	Note
Study	TransplantLines gut metagenome (PRJNA1047900)	Zhang et al. 2024, mSystems
Samples analysed today	6 total: 3 LTR + 3 HC	Same 6 samples, both Kraken2/Bracken and MetaPhlAn4
Subsampling depth	1,000,000 read pairs per sample	Uniform across all 6 — fair group comparison
Full-depth copies (not used today)	SRR27027652 (LTR), SRR27027756 (HC)	Set aside for homework / Day 5 mini-project
Soil sample	None in Day 3	Reserved dataset (PRJDB18822) for the mini-project — don't peek!

All 6 samples are gut — see data-README.md for full provenance and the subsampling rationale.

SECTION 1

QC & Preprocessing

FASTQC • MultiQC • fastp • Host Removal

Quality Control: FASTQC & MultiQC

FASTQC

Per-sample QC

Per-base quality

Phred scores along read length — look for quality drop at 3' end

Per-sequence quality

Distribution of mean quality per read

GC content

Should be approximately normal; spikes indicate contamination

Sequence duplication

High duplication → overamplification or low library complexity

Adapter content

Detect Illumina adapters that must be trimmed before analysis

Overrepresented sequences

Adapter dimers, ribosomal RNA, or contaminants

```
fastqc sample_R1.fastq.gz sample_R2.fastq.gz -o qc_output/
```

MultiQC

Aggregate across samples

- Aggregates FASTQC reports from all samples into one interactive HTML report
- Works with 100+ bioinformatics tools: FASTQC, Trimmomatic, Bowtie2, STAR, QUAST, Kraken2...
- Side-by-side comparison of quality metrics across many samples
- Instantly identify outlier samples (low quality, wrong GC content, etc.)
- Generates publication-quality figures (barplots, heatmaps, line plots)
- Supports custom modules via plugin system

```
multiqc qc_output/ -o multiqc_report/
```

Trimming & Host DNA Removal

fastp

Fast & feature-rich trimmer

- Adapter auto-detection (no need to specify adapter sequences)
- Quality trimming: --qualified_quality_phred 20
- Minimum read length: --length_required 50
- Deduplication: --dedup
- JSON + HTML report generated automatically
- Much faster than Trimmomatic (multi-threaded)

```
fastp -i R1.fastq.gz -I R2.fastq.gz \
  -o R1_trimmed.fastq.gz -O R2_trimmed.fastq.gz \
  -q 20 -l 50 --detect_adapter_for_pe -j fastp.json
```

Alternative trimmer: Trimmomatic (ILLUMINACLIP:adapters.fa:2:30:10 LEADING:3 TRAILING:3 SLIDINGWINDOW:4:15 MINLEN:36)
Beyond the host: reagent and lab-derived contamination (the "kitome") can dominate low-biomass samples — always include extraction/negative controls (Ch.2).

Host DNA Removal

Why? Human samples may contain 50–90% host DNA.
Without removal, host reads dominate assembly and classification, wasting compute and distorting results.

Strategy: Bowtie2 / BWA

1. Build index of host genome (GRCh38 for human)
2. Map reads to host: Bowtie2 --very-sensitive
3. Extract UNMAPPED reads → microbial reads
4. Convert SAM → BAM → filter unmapped

```
bowtie2 -x GRCh38 -1 R1.fastq.gz -2 R2.fastq.gz \
  --un-conc-gz clean_%.fastq.gz \
  -p 8 --very-sensitive | samtools view -bS > host.bam

# clean_1.fastq.gz and clean_2.fastq.gz = microbial reads
```

SECTION 2

Read-Based Taxonomy

Kraken2 • Bracken • MetaPhlAn4

Kraken2 & Bracken: k-mer Taxonomic Classification

How Kraken2 Works

1. Database build

k-mer extracted from all reference genomes, stored in hash table with taxon IDs

2. Read classification

Each read split into k-mers (default k=35). Each k-mer queried against database.

3. LCA assignment

Lowest Common Ancestor of all matching k-mer taxa assigned to read. Handles multi-mapping.

4. Output

Per-read classification + report with counts at each taxonomic level.

```
kraken2 --db /path/to/kraken2_db --paired R1.fastq.gz R2.fastq.gz --report kraken2.report --
output kraken2.out
bracken -d kraken2_db -i kraken2.report -o bracken.txt -r 150 -l S
```

Bracken

Bayesian Re-estimation of Abundance with KrakEN

Kraken2 classifies reads but gives counts, not true abundances. Bracken uses Bayesian re-estimation to provide accurate species-level abundance estimates from genus/species-level counts.

MetaPhlAn4

Marker gene based profiling. Uses ~5.1M clade-specific marker genes (not k-mers). More sensitive for novel species. Requires smaller database than Kraken2. Integrates with HUMAnN3.

SECTION 3

De Novo Assembly

MEGAHIT • SPAdes • Assembly QC

De Novo Assembly: Concepts

Goal: reconstruct longer genomic sequences (contigs) from short, overlapping reads — without a reference genome.

de Bruijn Graph (DBG)

Reads split into k-mers (overlapping subsequences of length k). Nodes = unique k-mers. Edges = overlaps. Contigs = paths through graph. Dominant approach for NGS.

k-mer Size

Smaller k: more connections, handles errors better, but more ambiguity.
Larger k: more specific, less ambiguity, but sensitive to errors.
MEGAHIT uses multiple k-mers iteratively.

Contigs vs Scaffolds

Contigs: continuous assembled sequences (no gaps).
Scaffolds: contigs joined using paired-end information with N-filled gaps.
For metagenomics, contigs ≥ 1000 bp are typically used downstream.

Coverage Depth

More reads → higher coverage → better assembly.
Recommended: ≥10× coverage per genome.
Abundant organisms assembled better than rare ones.
Very rare organisms may not assemble at all.

MEGAHIT vs metaSPAdes

Feature	MEGAHIT	metaSPAdes
Speed	★★★★★ Very fast (minutes–hours)	★★★☆☆ Slow (hours–days)
Memory	★★★★★ Low (8–32 GB typical)	★★☆☆☆ High (100+ GB for complex)
Assembly quality	★★★☆☆ Good	★★★★★ Better for complex/low cov.
k-mer strategy	Multiple k-mers iteratively	Multiple k-mers simultaneously
Long reads support	No	Yes (hybrid assembly)
Best for	Large datasets, resource-limited, standard QC	Complex communities, higher quality assemblies
Command	`megahit -1 R1.fq.gz -2 R2.fq.gz -o megahit_out`	`spades.py --meta -1 R1.fq.gz -2 R2.fq.gz -o spades_out`

For this course we use MEGAHIT (speed + memory). Use metaSPAdes for publication-quality assemblies.

Assembly QC: QUAST & SEQKIT

QUAST / MetaQUAST

Metric	Better	Description
Total length		Sum of all contig lengths — estimate of total metagenome size sequenced
# Contigs	lower	Total assembled sequences. Fewer contigs = more complete assembly
N50	higher	Contig length at which 50% of the assembly is in contigs of this length or longer
L50	lower	Fewest contigs whose combined length covers 50% of assembly. Fewer = better
Largest contig	higher	Longest single sequence — indicator of assembly continuity
GC content		Should match expected GC% for the community; flags contamination

```
quast.py assembly.fasta -o quast_output --threads 8
```

SEQKIT

Swiss-army knife for sequence files (FASTA/FASTQ).

`seqkit stats`
Summary: length, N50, count

`seqkit grep`
Extract sequences by name/pattern

`seqkit sort`
Sort by length (longest first)

`seqkit seq`
Filter contigs ≥ 1000 bp

`seqkit sample`
Subsample 10% of reads

`seqkit fx2tab`
Convert to tabular (name, len, GC)

AFTERNOON PRACTICAL

Today's Hands-On Session

1 QC & Trimming

Run FASTQC on shotgun reads. Check quality with MultiQC. Trim adapters with fastp. Compare before/after QC stats with SEQKIT.

2 Host Read Removal

Remove human reads using Bowtie2 against GRCh38. Count reads before/after. Inspect % host contamination across soil vs gut samples.

3 Taxonomic Classification

Run Kraken2 on cleaned reads. Apply Bracken for species-level abundance. Compare soil vs gut microbiome profiles.

4 De Novo Assembly

Assemble cleaned reads with MEGAHIT. Run QUAST for assembly statistics. Filter contigs ≥ 1000 bp with SEQKIT. Compare assemblies.

Day 3 Summary

Amplicon vs Shotgun	Shotgun sequences ALL DNA — gives taxonomy AND functional potential + enables MAG reconstruction.
QC	FASTQC identifies read quality issues; fastp trims adapters and low-quality bases efficiently.
Host removal	Map reads to host genome with Bowtie2; retain only unmapped (microbial) reads for downstream analysis.
Kraken2 + Bracken	k-mer classification provides fast taxonomy; Bracken re-estimates species-level abundance.
MEGAHIT	Fast, memory-efficient assembler using iterative de Bruijn graphs. Standard for metagenomics.
QUAST + SEQKIT	Evaluate assemblies (N50, contig count); manipulate FASTA/FASTQ with flexible SEQKIT toolkit.

Day 4 tomorrow → MAGs: Binning (MetaBAT2) → Quality (CheckM2) → Taxonomy (GTDB-Tk)

KEY REFERENCES

Wood et al. (2019)

Improved metagenomic analysis with Kraken 2

Genome Biology 20: 257

Lu et al. (2017)

Bracken: estimating species abundance in metagenomics data

PeerJ Computer Science 3: e104

Blanco-Míguez et al. (2023)

Extending metagenomic profiling with uncharacterized species using MetaPhlAn 4

Nature Biotechnology 41: 1633–1644

Li et al. (2015)

MEGAHIT: ultra-fast single-node solution for large metagenomics assembly

Bioinformatics 31: 1674–1676

Chen et al. (2018)

fastp: an ultra-fast all-in-one FASTQ preprocessor

Bioinformatics 34: i884–i890

Langmead & Salzberg (2012)

Fast gapped-read alignment with Bowtie 2

Nature Methods 9: 357–359

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